1. Executive Verdict

Final Acceptance Gate Classification: MATHEMATICAL REPAIR REQUIRED \to READY FOR PILOT ONLY (Strictly Conditional on the Repairs in Sections 4 and 6; Zero Empirical Claims Currently Validated).
An independent, hostile audit of the **PIGOU-J** proposal (and its predecessors **MORPH-KV**, **AER**, and **CORTEX**) against primary 2025–2026 literature, first-principles physics, queueing theory, and experimental statistics yields the following findings:

1. **Verdict on Predecessors (AER, CORTEX, MORPH-KV): PERMANENTLY REJECTED.**
    - **AER** relies on dimensionally inconsistent carbon equations, uncalibrated u^{0.7} utilization power curves, and token-proportional energy proxies that misprice batched requests by up to 158\%.
    - **CORTEX** relies on sequential small-to-large cascades that inflate tail latency, wide-area network (WAN) shifting of interactive queries that increases net latency and cold-KV recomputation energy, static split-conformal thresholds that break under workload drift, and in-band decode power caps (nvmlDeviceSetPowerManagementLimit) that fail to bind on memory-bound dense decode on H100 GPUs (which saturate 2.41 TB/s HBM2e bandwidth at \sim 350\text{ W} versus 700\text{ W} TDP).
    - **MORPH-KV** commits two fatal errors: (a) it confuses ex-post cooperative **Shapley energy attribution** (*JouleShare / JCalib*, arXiv:2608.00026) with ex-ante **counterfactual marginal dispatch cost**, and (b) it claims novelty over an un-isolatable 7-module assembly whose hardware and caching mechanisms already exist in 2026 preprints (**PowerSlider** arXiv:2608.21719, **Tri-serve** arXiv:2606.29629, **CacheRoute** arXiv:2608.19677, **Cognitora v0.9**, and **vLLM-SR** arXiv:2603.21354).
2. **Verdict on the Initial PIGOU-J Proposal: GENUINELY NOVEL CORE HYPOTHESIS, BUT CONTAINS THREE CRITICAL FLAWS REQUIRING IMMEDIATE REPAIR.**
    - **Flaw A — Prefill Power Double-Counting in \Delta \hat{E}_{\text{ext}} (Mathematical & Physical Error):** In the initial PIGOU-J equations, prefill self-energy was written as P_{\text{pre}}\Delta \tau_{\text{pre}} while the prefill externality simultaneously added \mathbb{I}\{n_w > 0\} P_{\text{dec}}(n_w, M_w)\Delta \tau_{\text{pre}}. A single GPU cannot simultaneously execute a dedicated prefill kernel drawing P_{\text{pre}} Watts *and* a decode kernel drawing P_{\text{dec}} Watts over the same time interval \Delta \tau_{\text{pre}}. Pausing n_w decode requests during a non-mixed prefill step shifts their completion time later by \Delta \tau_{\text{pre}} (imposing a **latency externality** n_w \Delta \tau_{\text{pre}}), but does *not* double instantaneous GPU board power.
    - **Flaw B — Invalid Two-Sided Equality Bound in IW-dACI Theorem 2 (Mathematical Proof Error):** In selective LLM routing (2026.acl-srw.70), when the miscoverage state \alpha_t \ge 1 routes all traffic to the small model (A_t = 1), the realized error \text{err}_t = A_t \mathbb{I}\{Y_t = 0\} is *not* deterministically equal to 1 (unlike standard conformal prediction where \alpha_t \ge 1 outputs the empty set \hat{C}_t = \emptyset), because the small model still answers correctly with probability \pi = \Pr(Y_t=1) and unaudited steps (I_t = 0) yield \widehat{\text{err}}_t = 0. Consequently, an unclipped \alpha_t can drift to +\infty, whereas clipping \alpha_t to an upper bound 1 + \gamma/p_{\text{audit}} introduces a one-sided

slack variable that breaks two-sided telescoping \vert{}\frac{1}{T}\sum \text{err}_t - \alpha\vert{} \le O(1/T) and establishes only a **one-sided upper risk bound** \frac{1}{T}\sum \text{err}_t \le \alpha + O(1/T).

- **Flaw C — Physically Impossible Live GPU Process Forking in the Pilot (Experimental Systems Error):** The proposed "200-intervention causal fork test" assumed live CUDA state on a running vLLM/SGLang worker could be instantaneously forked onto a shadow GPU pair to measure "exact" counterfactual energy. In reality, CUDA contexts and NVLink/NCCL handles cannot survive os.fork(), migrating 40–60 GB of KV cache takes hundreds of milliseconds, and NVML energy counters (nvmlDeviceGetTotalEnergyConsumption) update at 20\text{ to }100\text{ ms} PMU intervals. Counterfactual energy must instead be measured via **seeded deterministic busy-period trace-prefix replay** (following *JouleShare*) and treated as a **noisy empirical estimator** with quantified replay variance \sigma_{\text{replay}}^2, never as a zero-error constant.

2. Claim-by-Claim Audit (Requirement 1)

Every substantive mathematical, hardware, novelty, and experimental claim across PIGOU-J, MORPH-KV, AER, and CORTEX is audited below.

Claim ID	Source & Location	Exact Claim	Verification Method	Evidence & Primary Source	Confidence	Failure Implications	Status
C-01	AER / CAPS Eq. 1–3	Linear token proxy \hat{c}_i = \alpha_p n_i^{\text{in}} + \alpha_d \hat{n}_i^{\text{out}} and P(u)=P_{\text{idle}}+\Delta P u^{0.7} accurately price batched request energy.	Compare against vLLM counterfactual coalition replays on H100/A800/A40 GPUs.	Luo et al. (arXiv:2608.00026) show token-proportional attribution assigns 2,527.6\text{ J} to a long-prefill request whose continuous-batching Shapley value is 980.1\text{ J} (+158\% error).	High	Misroutes long-prompt requests and distorts multi-tenant energy accounting.	**Contradicted**
C-02	CORTEX / AER	Per-request spatial WAN shifting and temporal	Audit WAN RTT, KV-cache transfer volume,	arXiv:2610.13061 models inference as a rigid	High	Destroys TTFT SLOs and increases total cluster	**Contradicted**

Claim ID	Source & Location	Exact Claim	Verification Method	Evidence & Primary Source	Confidence	Failure Implications	Status
		deferral reduce interactive LLM carbon by 25–45%.	and MOER staleness sensitivity.	non-migratable load; arXiv:2407.02390 shows spatial shifting can increase emissions; arXiv:2602.10900 proves MOER staleness steers load into emission ramps.		Joules via cold remote prefill.	
C-03	CORTEX / MORPH-KV Phase 1	nvmlDeviceSetPowerManagementLimit is universally ineffective for memory-bound LLM decode.	Cross-check dense vs. MoE decode power-capping studies on H100/H200.	arXiv:2605.11999 proves caps fail to bind on memory-bound dense decode and H100 saturates 2.41 TB/s HBM2e at \sim 350\text{ W} (arXiv:2604.11391); however, arXiv:2609.11133 proves calibrated caps save 32.3\% on disaggregated MoE	High	Overstates failure of power capping if applied to wide-EP MoE decode without qualification.	**Partially Supported** (True for dense decode above 350 W on H100; false for calibrated MoE lanes)

Claim ID	Source & Location	Exact Claim	Verification Method	Evidence & Primary Source	Confidence	Failure Implications	Status
				decode lanes (5\times more than dense).			
C-04	MORPH-KV Novelty Claim 2	A 100 ms Vote-Commit nvmlDeviceSetGpuLockedClocks daemon locking decode to the 810 MHz knee and pacing 10% of prefill at 1800 MHz is a novel contribution of MORPH-KV.	Direct text and algorithmic comparison against 2026 arXiv preprints.	**PowerSlider** (arXiv:2608.21719, Alg. 1 & Fig. 4) introduces the 50–100 ms Vote-Commit DVFS daemon and 810 MHz A100 knee; **Tri-serve** (arXiv:2606.29629, Sec. III) introduces 10% prefill pacing at 1800 MHz and stall clock scaling; **AFlex** (arXiv:2608.01891) introduces operator-level Attention/FFN DVFS.	High	Immediate rejection for claiming published 2026 mechanisms as novel contributions.	**Contradicted**
C-05	MORPH-KV Stage 1C & 3D	Lease-bound kv_epoch BLAKE3 prefix	Direct comparison against 2026 orchestratio	BLAKE3 chains and etcd kv_epoch leases are	High	Rejection for prior-art collision; also *CacheRout*	**Contradicted**

Claim ID	Source & Location	Exact Claim	Verification Method	Evidence & Primary Source	Confidence	Failure Implications	Status
		tracking and k_b = \max(1, \lceil \lambda_b/q_{\text{cap}}\rceil) LPT prefix routing are novel to MORPH-KV.	n systems.	from **Cognitora v0.9**; k_b LPT sizing and 345 ms table rebuilds are from **CacheRoute** (arXiv:2608.19677, Alg. 1).		*e* proves prefix affinity hurts 32B models when KV work is small.	
C-06	MORPH-KV Novelty Claim 1	Dispatching requests to minimize predicted Shapley energy \phi_i^{\text{Shapley}}(B_w \cup \{i\}) (JCalib) minimizes marginal cluster energy.	First-principles game-theoretic and marginal calculus derivation.	By Shapley efficiency, \sum_{j \in B_w \cup \{i\}} \phi_j = E(B_w \cup \{i\}), which averages over all subsets S \subseteq B_w and charges newcomers a share of incumbents' sunk base energy E_0/(n_w+1), reversing optimal routing order on submodular curves (Theorem 1, Section 4).	High	Sistematically misroutes requests away from active batches with high sunk energy.	**Contradicted**
C-07	PIGOU-J	Pricing	Literature	**MARCS**	High	Latency-onl	**Partially**

Claim ID	Source & Location	Exact Claim	Verification Method	Evidence & Primary Source	Confidence	Failure Implications	Status
	(Turn 2) Novelty Claim	incumbent delay externality n_w \Delta \tau under continuous batching is a new routing formulation.	audit of 2026 continuous-batching schedulers.	(OpenReview:UEBPHGLSyO) already formulates Social Cost (SC) routing as arrival latency + incumbent-delay externality n_k[g(W_k +s, M_k) - g(W_k, M_k)]; **TAPER** (arXiv:2605.06914) formalizes branch externality E_t(k)=T(S(k))-T(S_0); **CADENCE** formalizes speculation externality.		y externality routing is already occupied by *MARCS*; novelty must reside strictly in **physical Joule externality** and its divergence from latency and Shapley cost.	**Supported** (Joule externality is open; latency externality is prior art)
C-08	PIGOU-J (Turn 2) Eq. 2 (\Delta \hat{E}_{\text{ext}})	During an uncached prefill of duration \Delta \tau_{\text{pre}}, incumbent energy externality is \mathbb{I}\{n_w > 0\} P_{\text{dec}}(n_w,	Physical power conservation audit over busy period [t_0, t_1].	**False.** A GPU executing a non-mixed prefill step draws P_{\text{pre}} (not P_{\text{pre}} + P_{\text{dec}}), while a mixed chunked-prefill step	High	Over-penalizes prefill on active workers by 150\text{ to }350\text{ W} \times \Delta \tau_{\text{pre}}.	**Contradicted (Repaired in Sec. 4)**

Claim ID	Source & Location	Exact Claim	Verification Method	Evidence & Primary Source	Confidence	Failure Implications	Status
		M_w)\Delta \tau_{\text{pre}} on top of \Delta \hat{E}_{\text{self}} = P_{\text{pre}}\Delta \tau_{\text{pre}}.		draws P_{\text{mixed}}(p_{\text{eff}}, n_w, M_w) *instead* of P_{\text{dec}}(n_w, M_w). Adding P_{\text{dec}}\Delta \tau_{\text{pre}} double-counts board power during prefill.			
C-09	PIGOU-J (Turn 2) Theorem 2 (IW-dACI)	Clipping \alpha_{t+1} to [-\gamma/p_{\text{audit}}, 1+\gamma/p_{\text{audit}}] yields a two-sided deterministic bound \left\vert{}\frac{1}{T-h}\sum \widehat{\text{err}}_t - \alpha\right\vert{} \le \frac{1+2\gamma/p_{\text{audit}}}{\gamma(T-h)}.	Algebraic telescoping verification under upper-bound clipping when \text{err}_t = A_t \mathbb{I}\{Y_t = 0\}.	**False for the lower tail.** When \alpha_t \ge 1, A_t = 1 does not force \widehat{\text{err}}_t = 1 (since Y_t can be 1 or I_t=0). Upper-clipping introduces a non-negative slack \xi_t \ge 0 in \alpha_{t+1} = \alpha_t + \gamma(\alpha - \widehat{\te	High	Claiming a two-sided bound is mathematically false whenever the small model's base error rate is strictly less than \alpha.	**Contradicted (Repaired to One-Sided Safety Bound in Sec. 4)**

Claim ID	Source & Location	Exact Claim	Verification Method	Evidence & Primary Source	Confidence	Failure Implications	Status
				xt{err}}_{t-h}) - \xi_t, so telescoping yields a **one-sided upper bound** \frac{1}{T-h}\sum \widehat{\text{err}}_t \le \alpha + O(1/T), not a two-sided equality.			
C-10	PIGOU-J (Turn 2) Sec. 12.1	Live CUDA worker state can be forked onto a shadow GPU pair at 200 decision epochs to measure "exact" counterfactual energy E(B_w \cup \{r_t\}) - E(B_w).	CUDA runtime & NVML telemetry architecture check.	CUDA contexts cannot be forked via os.fork(); migrating 40+ GB KV state perturbs timing; and nvmlDeviceGetTotalEnergyConsumption updates at 20\text{ to }100\text{ ms} PMU intervals with \pm 1\text{ to }5\% variance.	High	Experiment as originally written cannot be implemented on real GPUs and overclaims measurement exactness.	**Contradicted (Replaced with Seeded Prefix-Replay in Sec. 5)**
C-11	PIGOU-J (Turn 2) Sec. 12.1	N=12 paired-seed runs guarantee >90\% statistical	Paired Wilcoxon / t-distribution power derivation.	True *only if* \sigma_{\text{diff}} \le 2.5\%. Because \sigma_{\te	High	Under-powered experiment could produce false	**Unverified** (Requires Stage-0 AA Variance Pilot)

Claim ID	Source & Location	Exact Claim	Verification Method	Evidence & Primary Source	Confidence	Failure Implications	Status
		power to detect a 4\% Joule reduction assuming \sigma_{\text{diff}} = 2.5\%.		xt{diff}} has never been measured on the target cluster and thermal drift can induce 5\text{ to }15\% power variance on air-cooled nodes, \sigma_{\text{diff}} = 2.5\% is an unverified assumption.		negatives or noisy false positives.	
C-12	PIGOU-J (Turn 2) Hypothesis H_1	PIGOU-J reduces total node Joules per 1K goodput tokens by \ge 5\% (homogeneous) and \ge 8\% (two-tier) over MARCS-SC and JCalib-Shapley.	Requires physical multi-GPU testbed execution.	Zero testbed runs exist in the workspace.	High	Cannot be claimed as a result; may fail on homogeneous pools if power is nearly constant across active batch sizes n \ge 1.	**Unverified Hypothesis**

3. Verified Related-Work and Novelty Analysis (Requirement 2)

3.1 Comprehensive Comparison Against Closest 2025–2026 Systems

To establish whether *any* defensible novelty survives after stripping borrowed components, the table below compares PIGOU-J against the 11 closest verified systems in the 2025–2026 literature.

Competing System (Verified Source)	Core Mechanism & Optimization Target	Hardware & Serving Substrate	What Is Empirically Demonstrated	Exact Overlap with PIGOU-J	What Remains Genuinely Open for PIGOU-J
1. MARCS (OpenReview: UEBPHGLSyO, 2026)	Social Cost (SC) routing pricing self latency + **incumbent-delay externality** n_k[g(W_k+s, M_k) - g(W_k, M_k)] + guarded Thompson Sampling queue priority.	vLLM (mixed continuous batching & P/D disaggregated) + Vidur simulator.	-9.9\%/-1.9\% mean/P99 latency (mixed) and -12.3\%/-2.3\% (disaggregated) vs. SQF+FIFO across 20 seeds.	**Complete overlap on the latency-externality routing term** \Delta L_{\text{self}} + \beta_L n_w \Delta \tau and excess-cost decomposition with trajectory drift \Xi_T.	**MARCS is 100% energy-blind.** It treats step time g(W_k, M_k) as the sole cost, ignoring idle-to-active power jumps (P_{\text{idle}} \to P_{\text{act}}(1)), submodular HBM batch power amortization, and cross-tier Watt differences.
2. JouleShare & JCalib (Luo et al., arXiv:2608.00026, July 2026)	Offline 2^n coalition replay (n=8) for exact **Shapley active GPU energy** \phi_i^{\text{Shapley}}; JCalib regressor (0.003\text{ ms}).	Single-GPU vLLM on A40, A800, H100; models 1.5B–14B.	Token attribution overcharges long prompts by +158\% (2,527.6\text{ J} vs. 980.1\text{ J}); JCalib reaches 0.116 (static) and 0.177 (continuous) L1 error.	Uses counterfactual batch energy profiling on vLLM via NVML active power integration \int \max(P_S(t)-P_{\text{idle}}, 0)\,dt.	**JouleShare is strictly ex-post accounting**, not routing. More importantly, its Shapley target \phi_i divides sunk incumbent base energy rather than pricing the ex-ante marginal increment E(B \cup \{i\}) - E(B).
3. TAPER (Goel et al., arXiv:2605.06914, May 2026) & **CADENCE**	Per-step intra-request **branch externality** E_t(k)=T(S(k))-	SGLang on 8×A100-80GB serving Qwen3-32B with Azure	1.77\times goodput vs. IRP-OFF and 99\% SLO attainment vs.	Formalizes step-elongation externalities imposed on co-batched	Operates strictly **within a single batch step** to regulate

Competing System (Verified Source)	Core Mechanism & Optimization Target	Hardware & Serving Substrate	What Is Empirically Demonstrated	Exact Overlap with PIGOU-J	What Remains Genuinely Open for PIGOU-J
(ICLR'27 sub.)	T(S_0) (*TAPER*) and **speculation externality** (*CADENCE*) bounded against batch slack.	traces.	48\% for eager admission (0.3\text{ ms} planner overhead).	serial requests in continuous batching.	parallel branch/draft width; does not route requests across replicas/models or model GPU Joules.
4. PowerSlider (arXiv:2608.21719, Aug 2026)	Flex-SLO contracts (\alpha_c, \rho_c), KKT solver (PSOpt, <10\text{ ms}), and **50–100 ms Vote-Commit DVFS daemon** (nvmlDeviceSetGpuLockedClocks).	SGLang on A100-80GB; CodeLlama-34B & PTA reasoning traces.	78.3\% online goodput at -30\% power cap vs. 47.6\% baseline; characterizes 810\text{ MHz} A100 decode knee and P_{90}=4,050 reasoning tokens.	Pre-empts phase-aware SM clock locking, Vote-Commit DVFS, and Flex-SLO contracts.	Single-model cluster power-cap enforcement; notes that prediction-based routing breaks on heavy-tailed reasoning without mid-flight adaptation.
5. Tri-serve (arXiv:2606.29629, June 2026) & **AFlex** (Hu et al., arXiv:2608.01891, Aug 2026)	Stall-aware, AI-aware, and **10% thermal-paced prefill clock locking** (*Tri-serve*); **Attention–FFN disaggregation + operator DVFS** (*AFlex*).	vLLM v1 EngineCore (Qwen2-Omni); multi-node Azure traces (*AFlex*).	*Tri-serve*: 22\% energy/token reduction; *AFlex*: up to 49\% less J/token vs. DistServe (1.7\text{ J/tok} vs. 4.9\text{–}7.4\text{ J/tok} at 2 RPS).	Pre-empts phase/operator DVFS and highlights severe light-load (2 RPS) energy inefficiency (4.4\text{–}5.0\times higher J/tok than at 16 RPS).	Both optimize frequency scaling within a single model pipeline; neither prices inter-replica or inter-model counterfactual marginal Joules at dispatch.
6. PALS (Hankendi, Shahout, Yu, Coskun, arXiv:2605.21427, May 2026)	Offline power/throughput predictor + runtime feedback controller tuning (\text{power cap}, \text{batch	4×A100 NVLink nodes serving MoE models.	26.3\% efficiency improvement (95\% of oracle) and 4\text{–}7\times fewer QoS violations.	Explicitly defines the hierarchical split: *"the cluster decides where and how much, the node runtime decides how"*.	**PALS leaves the cluster "where and how much" routing layer unbuilt**—providing the exact architectural slot for PIGOU-J.

Competing System (Verified Source)	Core Mechanism & Optimization Target	Hardware & Serving Substrate	What Is Empirically Demonstrated	Exact Overlap with PIGOU-J	What Remains Genuinely Open for PIGOU-J
	size}) to maximize tok/J s.t. T(c) \ge T_{\text{target}}.				
7. CacheRoute (arXiv:2608.19677, Aug 2026) & **Cognitora v0.9** (Sep 2026)	Off-path LPT planner (345\text{ ms}) sizing k_b = \max(1, \lceil \lambda_b/q_{\text{cap}}\rceil) (*CacheRoute*); BLAKE3 + etcd kv_epoch leases (*Cognitora*).	30-destination 32B & 70B TP-2 deployments; Rust orchestrator over vLLM/SGLang.	Improves 70B serving, but **proves two 32B workloads suffer net loss** when affinity recovers too little KV work relative to induced load skew.	Pre-empts prefix-table planning and kv_epoch lease synchronization.	*CacheRoute* uses a static admission threshold and recommends offline shadow replay because it lacks an online **counterfactual Joule/latency trade-off price** between KV prefix reuse and queue skew!
8. vLLM-SR / WRP & The 1/W Law (Chen et al., arXiv:2603.21354, Mar 2026)	Workload–Router–Pool (WRP) framework, Matryoshka embeddings, 98\times faster CPU routing, and analytical **1/W Law** for context-length pool routing.	vLLM Semantic Router gateway + fleet simulators.	Shows context-length routing topology is a stronger energy lever than GPU generation when pools partition context lengths.	Overlaps with context-length / model-pool routing for energy efficiency.	The *1/W Law* is a static/analytical topology partitioning rule (arXiv:2603.21354), not a live batch-state-conditioned counterfactual marginal Joule dispatcher.
9. Proactive Conformal Gate (Uddin et al., 2026.acl-srw.70) & **DtACI** (Gibbs & Candès, JMLR 2024)	BGE logistic gate + split-conformal threshold + C-statistic diagnostic; online pinball-loss step-size expert	MMLU & GSM8K splits; non-exchangeable time-series.	87\% cost cut on MMLU, 35\% on GSM8K under exchangeability; proves code-llama-34b (C=10.92) is unroutable on MMLU.	Pre-empts proactive safety label Y=1-\mathbb{I}\{g\text{wrong}\}\mathbb{I}\{f\text{correct}\}, C-diagnostic, and	Does not address **selective bandit/censored feedback with h-step verification lag** when m_{\text{large}} is only queried

Competing System (Verified Source)	Core Mechanism & Optimization Target	Hardware & Serving Substrate	What Is Empirically Demonstrated	Exact Overlap with PIGOU-J	What Remains Genuinely Open for PIGOU-J
	aggregation (*DtACI*).			full-feedback ACI/DtACI.	on a p_{\text{audit}} Bernoulli subsample.

3.2 Strongest Rejection Arguments and What Genuinely Survives

1. **Strongest Rejection Argument Against PIGOU-J:***"Once you strip away PowerSlider's DVFS, CacheRoute's prefix planner, and Uddin et al.'s conformal gate, PIGOU-J is just **MARCS** (OpenReview:UEBPHGLSyO) with Watts multiplied onto the step duration function g(W_k, M_k). If GPU board power P_{\text{step}}(W_k, M_k) is nearly constant across active batch sizes n_k \ge 1, multiplying by a constant Watt factor does not change a single routing decision relative to MARCS."*
2. **Rigorous Rebuttal & Exact Condition Where PIGOU-J Is Substantively New:** The rejection argument is **valid** if PIGOU-J is evaluated only on homogeneous replicas at moderate-to-high steady load (n_w \ge 4 everywhere) without prefix caching or idle-consolidation! However, physical GPU power measurements in 2026 (arXiv:2608.01891, arXiv:2601.22362, arXiv:2608.00026, and cal23-gpupower.pdf) prove that marginal Joules \Delta E_{\text{marginal}} and marginal step delay \Delta L_{\text{social}} **strongly decouple and frequently point in opposite directions** in three concrete regimes:
    - **Regime 1: The Active-State Wake-Up Cliff (n_w = 0 \to 1 vs. n_w \ge 1 \to n_w + 1):** When a replica is idle (n_w = 0), its GPU sits at P_{\text{idle}} \approx 70\text{–}100\text{ W}. Admitting a single request (n_w = 0 \to 1) jumps GPU power to P_{\text{wall}}(1) \approx 220\text{–}300\text{ W} (an active jump P_{\text{act}}(1) \approx 150\text{–}200\text{ W}) because the GPU enters P0 frequency state and streams 100% of model weights from HBM on every decode step. Conversely, adding that same request to an *already active* replica (n_w = 2 \to 3) increases step time \tau_{\text{dec}} by < 2\% and dynamic power by < 10\text{ W}, because model weights are *already* being fetched from HBM for the 2 incumbents! **Latency-only MARCS routes the arrival to the idle replica (n_w = 0) to minimize latency externality (0\text{ ms} delay to incumbents), burning 150\text{–}200\text{ W} of unamortized active power. PIGOU-J packs the arrival into the active replica (n_w = 2) whenever latency slack permits (Z(t) small), eliminating the 4.4\text{–}5.0\times light-load Joule/token penalty documented in AFlex (arXiv:2608.01891)!**
    - **Regime 2: Prefix-Cache Hit vs. Queue-Load Trade-Off (*CacheRoute*'s Open Problem):** In *CacheRoute* (arXiv:2608.19677), the authors show that routing a request to a replica with a resident prefix cache hit saves prefill work, *but* if the prefix hit is small relative to the replica's queue depth, prefix affinity degrades performance. Because prefill is compute-bound (drawing 400\text{–}700\text{ W} on H100) while decode is memory-bound (\sim 300\text{–}350\text{ W}), PIGOU-J's exact marginal Joule equation prices the **watt-weighted prefill savings** \Delta E_{\text{KV}}(p_{t,w}^{\text{hit}}) directly against the **incumbent decode**

elongation externality \Delta E_{\text{ext}}(r_t \to B_w), solving *CacheRoute*'s trade-off dynamically per request without offline shadow replay!

- **Regime 3: Heterogeneous Model / GPU Pool Routing:** Across heterogeneous tiers (e.g., Qwen2.5-7B vs. Qwen2.5-72B, or A100 token decode vs. H100 prefill), latency and energy scale with completely different hardware constants (4.4\times energy gap per token).

4. Mathematical Derivations, Proofs, Counterexamples, and Proof Gaps (Requirement 3)

We now reconstruct the mathematics of PIGOU-J from first principles, repairing Flaws A and B identified in Section 1.

4.1 First-Principles Derivation of State-Conditioned Counterfactual Marginal Energy (\Delta E_{\text{marginal}})

System Horizon and State Definitions

Consider a cluster of W workers w \in \mathcal{W}. At decision time t=0 (shifting time origin to arrival r_t), worker w holds active batch B_w of size n_w = |B_w| with sorted remaining decode token requirements 0 < d_{w,(1)} \le d_{w,(2)} \le \dots \le d_{w,(n_w)}, queued uncached prefill tokens Q_w, and resident KV-cache token occupancy M_w. Let P_{\text{wall}, w}(\tau) [\text{W}] denote the physical board power of worker w at time \tau \ge 0, and let P_{\text{idle}, w} [\text{W}] be its steady-state idle power when no requests are active (n_w = 0). Over any fixed wall-clock horizon [0, H] sufficiently large to drain B_w \cup \{r_t\} (H \ge T_{\text{drain}}(B_w \cup \{r_t\})), the total physical energy consumed by worker w is:
E_{\text{wall}, w}(\mathcal{A}; [0, H]) = \int_0^H P_{\text{wall}, w}(\tau; \mathcal{A})\,d\tau = P_{\text{idle}, w} \cdot H + \int_0^{T_{\text{drain}}(\mathcal{A})} \!\! \big(P_{\text{wall}, w}(\tau; \mathcal{A}) - P_{\text{idle}, w} \big)\,d\tau = P_{\text{idle}, w} \cdot H + E_{\text{act}, w}(\mathcal{A})$$Because $P_{\text{idle}, w} \cdot H$ is invariant to whether r_t is placed on worker w (so long as the GPU is not powered off), the **exact counterfactual change in total cluster wall-clock energy over $[0, H]$** from placing r_t on worker w (action $\mathcal{A} = w$) versus not placing r_t on w (action $\mathcal{A} = \emptyset_w$) is identically equal to the change in active busy-period energy:$$\Delta E_{\text{marginal}}(r_t \mid s_w) \triangleq E_{\text{wall}, w}(B_w \cup \{r_t\}; [0, H]) - E_{\text{wall}, w}(B_w; [0, H]) = E_{\text{act}, w}(B_w \cup \{r_t\}) - E_{\text{act}, w}(B_w)
(Dimensional check: [\text{W}]\cdot[\text{s}] - [\text{W}]\cdot[\text{s}] = [\text{J}].)

Repairing Flaw A: Exact Step-Level Decomposition Under Continuous Batching (Colocated & Disaggregated)

In modern continuous batching engines (vLLM v0.6+/v1 and SGLang v0.4+), an arriving request r_t with uncached prefill length p_{t,w}^{\text{eff}} = p_t - p_{t,w}^{\text{hit}} and decode length d_t undergoes two phases:

1. **Phase 1: Prefill Execution (Mixed Chunked-Prefill vs. Dedicated Prefill):**
   - *Case 1a (Colocated Chunked Prefill — vLLM default):* If n_w > 0, r_t's uncached prefill of p_{t,w}^{\text{eff}} tokens is piggybacked alongside K_{\text{pre}} = \lceil p_{t,w}^{\text{eff}} / C_{\text{chunk}} \rceil ongoing decode steps of the n_w incumbents. Without r_t, those K_{\text{pre}} steps would have executed as pure

decode steps of duration K_{\text{pre}} \tau_{\text{dec}}(n_w, M_w) at active power P_{\text{dec}}(n_w, M_w). With r_t, they execute as mixed steps of total duration \tau_{\text{mix}}(p_{t,w}^{\text{eff}}, n_w, M_w^+) at active power P_{\text{mix}}(p_{t,w}^{\text{eff}}, n_w, M_w^+). Thus, the net counterfactual energy added during the prefill window is: \Delta E_{\text{pre}}^{\text{mixed}}(r_t \mid s_w) = P_{\text{mix}}(p_{t,w}^{\text{eff}}, n_w, M_w^+) \cdot \tau_{\text{mix}}(p_{t,w}^{\text{eff}}, n_w, M_w^+) - \mathbb{I}\{n_w > 0\} K_{\text{pre}} P_{\text{dec}}(n_w, M_w) \tau_{\text{dec}}(n_w, M_w) where \Delta \tau_{\text{pre-delay}}(r_t \mid s_w) = \tau_{\text{mix}}(p_{t,w}^{\text{eff}}, n_w, M_w^+) - \mathbb{I}\{n_w > 0\} K_{\text{pre}} \tau_{\text{dec}}(n_w, M_w) > 0 is the net step elongation imposed on each of the n_w co-resident requests!

- *Case 1b (Non-Mixed / Pause-Decode Prefill or Disaggregated Prefill Pool):* During \Delta \tau_{\text{pre}} = \tau_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w), only the prefill kernel runs on the prefill SMs, drawing active power P_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w). If colocated (pausing decode), the n_w decode requests are simply shifted later in time by \Delta \tau_{\text{pre}}: their subsequent decode steps still execute once, so the **added prefill energy is strictly P_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w) \cdot \Delta \tau_{\text{pre}} (zero double-counted P_{\text{dec}}\Delta \tau_{\text{pre}} power!)**, while the **latency externality** paid by the paused incumbents is n_w \cdot \Delta \tau_{\text{pre}}.

2. **Phase 2: Autoregressive Decode Execution (Capturing the Wake-Up Cliff and Incumbent Step Elongation):** Once r_t enters the decode batch, let N_{\text{ov}} = \min(\hat{d}_t, \tilde{d}_{w,\text{rem}}) be the overlap steps with the n_w incumbents and N_{\text{solo}} = \max(0, \hat{d}_t - N_{\text{ov}}) be the residual solo steps after the incumbents complete.
   - During each of the N_{\text{ov}} overlap steps (n_w \ge 1), baseline execution without r_t costs P_{\text{dec}}(n_w, M_w)\tau_{\text{dec}}(n_w, M_w) Joules/step, whereas co-batched execution with r_t costs P_{\text{dec}}(n_w+1, M_w^+)\tau_{\text{dec}}(n_w+1, M_w^+) Joules/step. Using the exact algebraic identity: P^+ \tau^+ - P^0 \tau^0 = \underbrace{(P^+ - P^0)\tau^0}_{\text{Self dynamic power }\delta E_{\text{self, dec}}} + \underbrace{P^+ (\tau^+ - \tau^0)}_{\text{Step-elongation externality }\delta E_{\text{ext, dec}}}
   - During each of the N_{\text{solo}} solo steps (including *all* \hat{d}_t steps when n_w = 0, i.e., **waking an idle worker**!), the worker transitions from n=0 (active power 0) to n=1 (active power P_{\text{dec}}(1, M_{\text{solo}}) = P_{\text{wake}, w} + P_{\text{dyn}}(1)), costing the full unamortized single-stream energy P_{\text{dec}}(1, M_{\text{solo}})\tau_{\text{dec}}(1, M_{\text{solo}}) per step!

Combining Phases 1 and 2 yields the **Repaired, Conservation-Consistent PIGOU-J Marginal Energy Equations**:

\Delta \hat{E}_{\text{self}}(r_t \mid s_w) = P_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w) \tau_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w) + N_{\text{ov}} \big(P_{\text{dec}}(n_w+1, M_w^+) - P_{\text{dec}}(n_w, M_w) \big) \tau_{\text{dec}}(n_w, M_w) + N_{\text{solo}} P_{\text{dec}}(1, p_t+\hat{d}_t) \tau_{\text{dec}}(1, p_t+\hat{d}_t) \Delta \hat{E}_{\text{ext}}(r_t \to B_w \mid s_w) = \mathbb{I}\{\text{chunk_mix} \wedge n_w > 0\} \Delta E_{\text{mix_ovhd}}(p_{t,w}^{\text{eff}}, n_w, M_w) + N_{\text{ov}} P_{\text{dec}}(n_w+1, M_w^+) \big(\tau_{\text{dec}}(n_w+1, M_w^+) - \tau_{\text{dec}}(n_w, M_w) \big)

$where \Delta E_{mix_ovhd} = P_{mix}\tau_{mix} - P_{pre}\tau_{pre} - K_{pre}P_{dec}\tau_{dec} \$ is zero under pause-decode or disaggregated prefill, and

\Delta \hat{E}_{\text{self}}(r_t \mid s_w) + \Delta \hat{E}_{\text{ext}}(r_t \to B_w \mid s_w) \equiv \widehat{E}_{\text{act}, w}(B_w \cup \{r_t\}) - \widehat{E}_{\text{act}, w}(B_w) = \Delta \hat{E}_{\text{marginal}}(r_t \mid s_w)
(Exact conservation verified: the sum of private marginal energy and Pigouvian externality energy equals the exact counterfactual change in worker active energy, with zero double-counting.)

4.2 Complete Proof of Theorem 1: Non-Equivalence and Rank-Reversal Between Shapley Attribution (JCalib) and Counterfactual Marginal Cost (PIGOU-J)

Theorem 1 (Incompatibility of Ex-Post Shapley Attribution with Optimal Ex-Ante Marginal Dispatch): *Consider a cluster of W \ge 2 workers serving requests over a single batch horizon. Let E_{\text{act}, w}(S) denote the active energy of serving subset S on worker w with E_{\text{act}, w}(\emptyset) = 0. Suppose a newcomer r_t arrives when worker w holds incumbent set B_w of size n_w = |B_w|.*

1. *(Structural Identity): The ex-post Shapley attribution \phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) equals the ex-ante counterfactual marginal cost \Delta E_{\text{marginal}}(r_t \mid B_w) = E_{\text{act}, w}(B_w \cup \{r_t\}) - E_{\text{act}, w}(B_w) for all incumbent configurations B_w if and only if E_{\text{act}, w}(S) is strictly additive across requests (E_{\text{act}, w}(S) = \sum_{j \in S} E_{\text{act}, w}(\{j\})), i.e., batching has zero energy synergy and zero congestion externality.*
2. *(Submodular Wake-Up Distortion): Suppose active batch energy follows the affine-wake-up law E_{\text{act}, w}(S) = E_{\text{wake}, w}\mathbb{I}\{S \neq \emptyset\} + \sum_{j \in S} e_{w,j} with wake-up energy E_{\text{wake}, w} > 0 and incremental energy e_{w,j} > 0. Then for every n_w \ge 1:* \Delta E_{\text{marginal}}(r_t \mid B_w) = e_{w, r_t}, \qquad \text{whereas} \qquad \phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) = \frac{E_{\text{wake}, w}}{n_w + 1} + e_{w, r_t} > \Delta E_{\text{marginal}}(r_t \mid B_w)
3. *(Strict Suboptimality of Shapley Dispatch): Under the affine-wake-up law above, for any two active workers w_1, w_2 (n_1, n_2 \ge 1) where e_{w_1, r_t} < e_{w_2, r_t} (worker w_1 has lower true marginal energy, e.g., due to a KV-cache prefix hit) but \frac{E_{\text{wake}, w_1}}{n_1 + 1} - \frac{E_{\text{wake}, w_2}}{n_2 + 1} > e_{w_2, r_t} - e_{w_1, r_t} (worker w_2 has a larger incumbent batch size n_2 > n_1 or smaller sunk wake-up cost), Shapley dispatch (JCalib) selects w_2, strictly increasing physical cluster energy consumption by e_{w_2, r_t} - e_{w_1, r_t} > 0 Joules.*

Complete Proof:

1. By definition of the Shapley value: \phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) = \sum_{S \subseteq B_w} \frac{|S|!(n_w - |S|)!}{(n_w + 1)!} \big(E_{\text{act}, w}(S \cup \{r_t\}) - E_{\text{act}, w}(S) \big) Because the weights \alpha_S = \frac{|S|!(n_w - |S|)!}{(n_w + 1)!} sum to 1 over S \subseteq B_w, \phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) = E_{\text{act}, w}(B_w \cup \{r_t\}) - E_{\text{act}, w}(B_w) holds for all B_w and r_t if and only if the marginal contribution D_{r_t}(S) \triangleq E_{\text{act}, w}(S \cup \{r_t\}) - E_{\text{act}, w}(S) is independent of S (so D_{r_t}(S) = D_{r_t}(\emptyset) = E_{\text{act}, w}(\{r_t\}) for all S \subseteq B_w). By induction on |S|, E_{\text{act}, w}(S \cup \{r_t\}) = E_{\text{act}, w}(S) + E_{\text{act}, w}(\{r_t\}) for all S, r_t is equivalent to E_{\text{act}, w}(S) = \sum_{j \in S} E_{\text{act}, w}(\{j\}).
2. Under E_{\text{act}, w}(S) = E_{\text{wake}, w}\mathbb{I}\{S \neq \emptyset\} + \sum_{j \in S} e_{w,j}, for any non-empty B_w (n_w \ge 1), both B_w and B_w \cup \{r_t\} are non-empty, so \mathbb{I}\{B_w \cup \{r_t\} \neq \emptyset\} - \mathbb{I}\{B_w \neq

\emptyset\} = 1 - 1 = 0. Hence \Delta E_{\text{marginal}}(r_t \mid B_w) = e_{w, r_t}. In the Shapley sum over S \subseteq B_w, the term S = \emptyset has weight \frac{0!\,n_w!}{(n_w+1)!} = \frac{1}{n_w+1} and marginal increment E_{\text{act}, w}(\{r_t\}) - E_{\text{act}, w}(\emptyset) = E_{\text{wake}, w} + e_{w, r_t}, while all 2^{n_w}-1 non-empty subsets S \neq \emptyset have marginal increment e_{w, r_t}. Summing gives \phi_{r_t}^{\text{Shapley}}(B_w \cup \{r_t\}) = \frac{E_{\text{wake}, w}}{n_w + 1} + e_{w, r_t}.

3. Total cluster active energy is E_{\text{cluster}}(w) = \sum_{k \neq w} E_{\text{act}, k}(B_k) + E_{\text{act}, w}(B_w \cup \{r_t\}) = \sum_{k=1}^W E_{\text{act}, k}(B_k) + \Delta E_{\text{marginal}}(r_t \mid B_w). Since \sum_{k=1}^W E_{\text{act}, k}(B_k) is constant with respect to the choice of w, minimizing E_{\text{cluster}}(w) is equivalent to minimizing \Delta E_{\text{marginal}}(r_t \mid B_w) = e_{w, r_t}, which uniquely selects w_1. However, under the stated condition \frac{E_{\text{wake}, w_1}}{n_1+1} + e_{w_1, r_t} > \frac{E_{\text{wake}, w_2}}{n_2+1} + e_{w_2, r_t}, we have \phi_{r_t}^{\text{Shapley}}(B_{w_1} \cup \{r_t\}) > \phi_{r_t}^{\text{Shapley}}(B_{w_2} \cup \{r_t\}), so Shapley dispatch selects w_2 and wastes e_{w_2, r_t} - e_{w_1, r_t} > 0 Joules. \blacksquare

4.3 Repairing Flaw B: Complete Proof of Theorem 2 (One-Sided Non-Exchangeable Safety Guarantee of Clipped IW-dACI)

To fix Flaw B (where \alpha_t \ge 1 could drift to +\infty without clipping, and upper clipping broke two-sided telescoping), we formalize **One-Sided Clipped Importance-Weighted Delayed ACI (IW-dACI)** and prove its exact one-sided safety bound.

Theorem 2 (Distribution-Free One-Sided Risk Control Under Clipped Importance-Weighted Delayed Auditing): *Let \{(x_t, Y_t)\}_{t=1}^T be any arbitrary, deterministic or stochastic, non-exchangeable sequence of prompts and ground-truth capability labels Y_t = 1 - \mathbb{I}\{m_{\text{small}}\text{ wrong}\}\mathbb{I}\{m_{\text{large}}\text{ correct}\} \in \{0, 1\}. At each step t, let the routing gate select A_t = \mathbb{I}\{\hat{s}(x_t) \ge \tau_t\} \in \{0, 1\} with threshold \tau_t = \text{Quantile}(1 - \Pi_{[0,1]}(\alpha_t), \mathcal{S}_{\text{cal}}) under the boundary convention \tau_t = +\infty \implies A_t = 0 whenever \alpha_t \le 0. Let I_t \sim \text{Bernoulli}(p_{\text{audit}}) (p_{\text{audit}} \in (0, 1]) be drawn independently of (x_t, Y_t, \mathcal{H}_t), revealing the true unsafe deferral indicator \text{err}_t = A_t \mathbb{I}\{Y_t = 0\} \in \{0, 1\} at step t + h (h \ge 0). Define the Horvitz-Thompson estimator \widehat{\text{err}}_t = \frac{I_t}{p_{\text{audit}}}\text{err}_t \in \{0, 1/p_{\text{audit}}\} (with \widehat{\text{err}}_s \triangleq \alpha for s \le 0) and the upper-clipped delayed recursion with \alpha_1 \in [0, 1]:*

$$\alpha_{t+1} = \min\!\Big(1,\; \alpha_t + \gamma \left(\alpha - \widehat{\text{err}}_{t-h} \right) \Big)$$

Then for every sample path and every horizon T > h:

1. *(Bounded State Invariant): For all t \ge 1, \alpha_t \in \left[-\frac{(h+1)\gamma}{p_{\text{audit}}},\; 1 \right].*
2. *(Deterministic One-Sided Sample-Path Safety Bound on Audited Estimator):* $\frac{1}{T - h} \sum_{t=1}^{T - h} \widehat{\text{err}}_t \le \alpha + \frac{1 + (h + 1)\gamma / p_{\text{audit}}}{\gamma (T - h)}$
3. *(High-Probability One-Sided Population Safety Bound on All Routed Traffic): With probability at least 1 - \delta over the independent Bernoulli audit coins \{I_t\}_{t=1}^{T-h} (conditioned on any arbitrary realization of prompts, queues, and labels \{(x_t,*

Y_t)\}_{t=1}^T), the true population unsafe deferral rate satisfies: $\frac{1}{T - h} \sum_{t=1}^{T - h} \text{err}_t \le \alpha + \frac{1 + (h + 1)\gamma / p_{\text{audit}}}{\gamma (T - h)} + \frac{1}{p_{\text{audit}}} \sqrt{\frac{\log(1/\delta)}{2 (T - h)}}$

Complete Proof:

1. **Bounded State Invariant:** The upper bound $\alpha_t \le 1$ holds for all $t \ge 1$ directly from the $\min(1, \cdot)$ operator and $\alpha_1 \le 1$. For the lower bound, observe that whenever $\alpha_s \le 0$ at any step s, the boundary convention enforces $\tau_s = +\infty \implies A_s = 0 \implies \text{err}_s = 0 \implies \widehat{\text{err}}_s = 0$. Suppose $\alpha_{t_0} > 0$ is the last positive state before an excursion into $(\infty, 0]$, so $\alpha_{t_0+1} \le 0$. Because $\widehat{\text{err}}_s \le 1/p_{\text{audit}}$ and $\alpha > 0$, any single step can decrease α by at most $\gamma(\widehat{\text{err}} - \alpha) < \gamma / p_{\text{audit}}$, so $\alpha_{t_0+1} > -\gamma / p_{\text{audit}}$. For $t \ge t_0 + 1$, so long as α remains ≤ 0, every new step $s \ge t_0 + 1$ produces $\widehat{\text{err}}_s = 0$. At most h in-flight delayed errors $\widehat{\text{err}}_{s}$ generated at steps $s \in \{t_0 - h + 1, \dots, t_0\}$ (when $\alpha_s > 0$) can arrive during steps $t \in \{t_0 + 1, \dots, t_0 + h\}$, each reducing α by at most $\gamma / p_{\text{audit}}$. After step $t_0 + h$, all arriving feedback corresponds to steps $s \ge t_0 + 1$ where $\widehat{\text{err}}_s = 0$, causing α to strictly increase by $+\gamma \alpha > 0$ per step until re-entering $(0, 1]$. Thus $\alpha_t \ge -\frac{(h+1)\gamma}{p_{\text{audit}}}$ for all $t \ge 1$.
2. **One-Sided Telescoping Inequality:** Define the non-negative upper-clipping slack $\xi_t \triangleq \max\!\big(0,\, \alpha_t + \gamma(\alpha - \widehat{\text{err}}_{t-h}) - 1\big) \ge 0$. Then for every $t \ge h + 1$: $\alpha_{t+1} - \alpha_t = \gamma \left(\alpha - \widehat{\text{err}}_{t-h} \right) - \xi_t \le \gamma \left(\alpha - \widehat{\text{err}}_{t-h} \right)$ Summing from $t = h + 1$ to T and indexing $s = t - h \in \{1, \dots, T - h\}$ yields: $\alpha_{T+1} - \alpha_{h+1} \le \gamma (T - h)\alpha - \gamma \sum_{s=1}^{T-h} \widehat{\text{err}}_s \implies \frac{1}{T - h} \sum_{s=1}^{T-h} \widehat{\text{err}}_s \le \alpha + \frac{\alpha_{h+1} - \alpha_{T+1}}{\gamma (T - h)}$ Since $\alpha_{h+1} \le 1$ and $\alpha_{T+1} \ge -\frac{(h+1)\gamma}{p_{\text{audit}}}$, we have $\alpha_{h+1} - \alpha_{T+1} \le 1 + \frac{(h+1)\gamma}{p_{\text{audit}}}$, proving Claim 2.
3. **Martingale Concentration Over Independent Audit Coins:** Let $\mathcal{F}_s$ be the σ-algebra generated by all system events up to the routing decision A_s at step s (and all audit coins $I_1, \dots, I_{s-1}$). Because A_s and Y_s are $\mathcal{F}_s$-measurable ($\text{err}_s \in \mathcal{F}_s$) and $I_s \sim \text{Bernoulli}(p_{\text{audit}})$ is drawn independently of $\mathcal{F}_s$, the conditional expectation is $\mathbb{E}[\widehat{\text{err}}_s \mid \mathcal{F}_s] = \frac{\mathbb{E}[I_s \mid \mathcal{F}_s]}{p_{\text{audit}}} \text{err}_s = \text{err}_s$. Thus $D_s \triangleq \text{err}_s - \widehat{\text{err}}_s$ is a martingale difference sequence with $\mathbb{E}[D_s \mid \mathcal{F}_s] = 0$. Since $\text{err}_s \in \{0, 1\}$ and $\widehat{\text{err}}_s \in \{0, 1/p_{\text{audit}}\}$, each difference D_s lies within an interval of length $L_s \le \frac{1}{p_{\text{audit}}}$ (specifically, if $\text{err}_s = 0$, $D_s = 0$ has range 0; if $\text{err}_s = 1$, $D_s = 1 - I_s/p_{\text{audit}} \in [1 - 1/p_{\text{audit}}, 1]$ has range $1/p_{\text{audit}}$). By the one-sided Azuma-Hoeffding inequality: $\Pr\!\left(\frac{1}{T - h} \sum_{s=1}^{T - h} \left(\text{err}_s - \widehat{\text{err}}_s \right) > \epsilon \right) \le \exp\!\left(-\frac{2 (T - h)^2 \epsilon^2}{(T - h) (1 / p_{\text{audit}})^2} \right) = \exp\!\left(-2 (T - h) p_{\text{audit}}^2 \epsilon^2 \right)$ Setting the right-hand side to δ, solving for $\epsilon = \frac{1}{p_{\text{audit}}}\sqrt{\frac{\log(1/\delta)}{2(T-h)}}$, and combining with

Claim 2 establishes Claim 3. \blacksquare

- **What Theorem 2 Does NOT Guarantee:** (1) It bounds the *average* unsafe deferral rate over T - h requests, not the instant conditional probability \Pr(Y_t = 0 \mid x_t) on a single adversarial prompt; (2) during the first h requests after an instantaneous regime shift, arrived requests use thresholds updated prior to the shift, incurring an unavoidable transient lag bounded by \frac{(h+1)}{T-h}; and (3) setting p_{\text{audit}} too small (e.g., p_{\text{audit}} < 0.02) inflates the Azuma-Hoeffding variance term by 1/p_{\text{audit}}.

5. Experiment Feasibility and Statistical-Power Audit (Requirements 4 & 5)

We now audit every proposed experiment to verify whether it is physically executable on real GPUs, whether the energy instrumentation resolution supports the estimand, and what sample size is mathematically required to achieve statistical power.

5.1 Four Levels of Scientific Inference (Strictly Separated)

Per Requirement 5, we separate the experimental hierarchy into four distinct claims that must not be conflated:

1. **Level 1 (Better Counterfactual Energy Prediction):** Does \Delta \hat{E}_{\text{marginal}}(r_t \mid s_w) predict the true busy-period active energy increment E_{\text{act}, w}(B_w \cup \{r_t\}) - E_{\text{act}, w}(B_w) with lower Mean Absolute Percentage Error (MAPE) and lower pairwise ranking inversion rate than Token-Proportional (e_m(p_{t,w}^{\text{eff}}+\hat{d}_t)) and JCalib-Shapley (\hat{\phi}_{r_t}^{\text{Shapley}})?
2. **Level 2 (Better Single-Step Routing Decisions):** At a fixed set of sampled cluster states \{s_w(t_k)\}_{w=1}^W, does selecting w^* = \arg\min_w \text{Score}_{\text{PIGOU-J}}(r_{t_k} \mid s_w) yield lower realized cluster busy-period energy than the replica selected by SQF, Token-Proportional, JCalib-Shapley, or MARCS-SC?
3. **Level 3 (Lower End-to-End Measured Node Energy at Matched Throughput, Latency, and Quality):** Over a continuous 15-minute steady-state trace replay at matched arrival rate \lambda, does PIGOU-J reduce **Total Node Joules per 1,000 Goodput Tokens** (\text{J/1K goodput tok}, including idle GPU power, CPU/DRAM RAPL power, and any shadow-audit GPU power) without increasing P99 TTFT, P99 TPOT, or quality risk beyond \alpha?
4. **Level 4 (Generalization):** Does the Level 3 improvement persist across (a) unseen prompt/decode length distributions (e.g., reasoning traces with heavy-tailed outputs), (b) colocated vs. disaggregated serving, and (c) different GPU architectures (A100 vs. H100)?

5.2 Audit of the 7 Proposed Experiments and Repair of Flaw C

Experiment Name	Flaw Identified in Previous Design	Repaired Executable Design & Estimand	Independent Experimental Unit	Instrumentation & Resolution Check	Cheaper Preliminary Test (Before Full Run)
Exp 0: Stage-0 AA Variance & Sensor Calibration Pilot *(Newly Mandated)*	Previous design *assumed* \sigma_{\text{diff}} = 2.5\% without measurement.	Run N_0 = 6 back-to-back paired replays of the *exact same* baseline policy (SQF) on matched seeds after thermal reset (\le 38^\circ\text{C}), comparing NVML cumulative mJ + RAPL against external rack PDU / IPMI. **Estimand:** Run-to-run paired SD \hat{\sigma}_{\text{diff}} and PDU calibration factor \kappa_g.	1 complete 15-min trace replay run (n = 6 pairs).	Over 15 min (900\text{ s}) at \sim 1,000\text{ W}, total energy is \sim 900\text{ kJ}; NVML 20\text{–}100\text{ ms} counter resolution (\sim 20\text{ J} at edges) induces < 0.005\% boundary quantization error.	3-minute mini-AA run (takes 30 minutes total) to verify thermal reset stability.
Exp 1: Offline Step-Surface Profiler (pigou-profiler)	Assumed single pass suffices; ignored thermal state drift during profiling.	Sweep (n \in \{0,1,2,4,8,16,32,64,128\},\, p \in \{0,64,256,1024,4096\},\, M \in \{10\%, 50\%, 85\%\}) for 100 steps per grid point (\ge 3\text{ s} duration per point) at locked thermal fan/temp state. **Estimand:** \tau_{\text{pre}}, P_{\text{pre}}, \tau_{\text{dec}}	1 sustained 100-step micro-batch execution (\times 5 repeats per grid point).	\ge 3\text{ s} window is \ge 30\times the 100 ms NVML PMU averaging window, eliminating aliasing.	Profile 1D decode slice (p=0, n \in \{0,1,4,16,64\}) in 5 minutes to measure P_{\text{wake}} and decode power slope.

Experiment Name	Flaw Identified in Previous Design	Repaired Executable Design & Estimand	Independent Experimental Unit	Instrumentation & Resolution Check	Cheaper Preliminary Test (Before Full Run)
		}, P_{\text{dec}}, P_{\text{idle}}.			
Exp 2: Seeded Busy-Period Prefix-Replay Test *(Replaces Impossible "Live Fork" Test)*	**Flaw C:** Live OS/CUDA forking of vLLM workers is physically impossible; single-step energy is below NVML 100 ms resolution.	Construct M = 100 deterministic multi-request batch prefixes (B_{w_1}, B_{w_2}) of 2\text{ to }6\text{ s} duration (temperature=0, fixed seed, Following *JouleShare*). For each prefix k, run 3 clean busy-period replays (\times R=5 repeats each to average out kernel jitter): (i) Baseline (B_{w_1}, B_{w_2}), (ii) Perturbation 1 (B_{w_1}\cup\{r_k\}, B_{w_2}), (iii) Perturbation 2 (B_{w_1}, B_{w_2}\cup\{r_k\}). **Estimand:** Level 1 MAPE & Level 2 Pairwise Decision Regret (in Joules).	1 deterministic batch-prefix scenario k \in \{1, \dots, 100\} (averaged over R=5 hardware replays).	Each busy-period replay lasts 2\text{ to }6\text{ s} (\sim 500\text{ to }2,000\text{ J}), well above NVML millijoule counter granularity. Standard error of \widehat{\Delta E}_{\text{replay}} is \sigma_{\text{rep}}/\sqrt{5}.	Run M=10 two-worker scenarios mirroring Theorem 1 (n_1=1 with prefix hit vs. n_2=4 cold) in 15 minutes.

Experiment Name	Flaw Identified in Previous Design	Repaired Executable Design & Estimand	Independent Experimental Unit	Instrumentation & Resolution Check	Cheaper Preliminary Test (Before Full Run)
Exp 3: Homogeneous 4-GPU Routing Pilot	Tested only 75\% load (where all GPUs are active n_w \ge 2 and power is flat, risking collinearity with MARCS).	Evaluate two distinct load points across N paired-seed 15-min runs: (a) **Light/Diurnal Load (\rho = 0.30, testing n_w = 0 \to 1 wake-up consolidation vs. MARCS spreading)**, and (b) **Moderate-High Mixed Prefix Load (\rho = 0.70, testing KV-hit prefill Watts vs. queue delay)**. **Primary Endpoint:** Node \text{J/1K goodput tok}.	1 complete 15-minute steady-state trace replay run per policy per seed.	Cumulative NVML mJ + CPU/DRAM RAPL over 900 s steady-state window; boundary-straddling requests accounted via Section 6 conservation test.	2-seed pilot at \rho = 0.30 and \rho = 0.70 comparing only PIGOU-J, MARCS-SC, and JCalib-Shapley.
Exp 4: Two-Tier Heterogeneous Model/GPU Evaluation	Did not isolate model-routing savings from replica-routing savings.	4-to-8 GPU pool pairing Qwen2.5-7B + Qwen2.5-72B-FP8 (or A100 + H100) under BurstGPT + LiveBench/LiveCodeBench/MMLU-Pro. Compares PIGOU-J + IW-dACI against Static Conformal + Token Cost and MARCS +	1 complete 15-min trace replay run.	Includes physical GPU energy of both tiers + shadow audit executions + host CPU RAPL.	Offline trace simulation using measured Exp 1 energy tables and exact benchmark correctness labels Y_t before full GPU replay.

Experiment Name	Flaw Identified in Previous Design	Repaired Executable Design & Estimand	Independent Experimental Unit	Instrumentation & Resolution Check	Cheaper Preliminary Test (Before Full Run)
		IW-dACI. **Primary Endpoint:** Node \text{J/1K goodput tok} (including shadow-audit Joules!) at empirical risk \le \alpha.			
Exp 5: Non-Stationary Drift & Selective Audit Test	Did not test adversarial burst-correlated difficulty where trough-auditing fails.	4-phase drift trace (Easy Chat \to Hard Code During 3x Arrival Burst \to Math Reasoning \to Mixed) comparing Static Split-Conformal, Trough-Audited Q-DtACI, and Bernoulli IW-dACI (p_{\text{audit}}=0.05). **Primary Endpoint:** Max rolling 200-query excess risk \max_t (\widehat{\text{Risk}}_{200}(t) - \alpha).	1 seeded 2,000-request stream realization (N=20 seeds).	Uses exact deterministic verification labels on LiveCodeBench and MATH-500.	Fast CPU-only trace replay using pre-recorded model outputs Y_t across 100 random seeds (completes in < 2 minutes!).
Exp 6: Energy Conservation & Reconciliation Audit	Treated as a post-hoc check rather than a hard validity gate.	Across all Exp 3 & Exp 4 runs, compare sum of attributed request energy + measured	Every 15-min experimental run.	Must satisfy $	\epsilon_{\text{rec}}

Experiment Name	Flaw Identified in Previous Design	Repaired Executable Design & Estimand	Independent Experimental Unit	Instrumentation & Resolution Check	Cheaper Preliminary Test (Before Full Run)
		P_{\text{idle}}\Delta T + straddle energy + daemon RAPL energy against total wall-clock NVML + RAPL + PDU energy. **Endpoint:** Relative reconciliation residual \epsilon_{\text{rec}}.			

5.3 Mathematical Sample-Size and Statistical-Power Derivation

Let $Y_{s, \pi}$ denote the primary endpoint (**Total Node Joules per 1,000 Goodput Tokens**) on seed $s \in \{1, \dots, N\}$ under policy π, and let $D_s = \frac{Y_{s, \text{base}} - Y_{s, \text{PIGOU-J}}}{Y_{s, \text{base}}}$ be the paired relative energy reduction on seed s. Let δ_{min} be the **Minimum Practically Meaningful Effect (MPME)**:

- Because NVML onboard power sensors exhibit $\pm 1\text{ to }2\%$ repeatability noise even after PDU calibration and simple node-level knobs require $\ge 4\text{ to }5\%$ gains to justify operational adoption, we define $\mathbf{\delta_{\text{min}} = 4.0\%}$ for single-model homogeneous routing (Exp 3) and $\mathbf{\delta_{\text{min}} = 7.0\%}$ for two-tier heterogeneous routing (Exp 4).
- For a two-sided paired Wilcoxon signed-rank test (Asymptotic Relative Efficiency $\text{ARE} \ge 0.955$ for normal errors and $\ge \frac{108}{125} = 0.864$ for all continuous symmetric distributions) controlling family-wise error rate at $\alpha_{\text{FWER}} = 0.05$ across $K_{\text{comp}} = 4$ primary baselines via Holm-Bonferroni (strictest per-test level $\alpha_1 = 0.05 / 4 = 0.0125$), the required number of independent paired-seed runs N for power $1 - \beta = 0.90$ as a function of the empirical paired standard deviation $\hat{\sigma}_{\text{diff}}$ measured in **Exp 0** is: $$N_{\text{req}}(\hat{\sigma}_{\text{diff}}) = \left\lceil \frac{1}{\text{ARE}_{\text{min}}} \left(\frac{\big(t_{N-1,\, 1 - \alpha_1/2} + t_{N-1,\, 1 - \beta} \big) \hat{\sigma}_{\text{diff}}}{\delta_{\text{min}}} \right)^{\!2} \right\rceil$$ Evaluating this exact formula for $\delta_{\text{min}} = 4.0\%$, $\alpha_1 = 0.0125$, $1-\beta = 0.90$, and $\text{ARE} = 0.955$:
  - If Exp 0 measures $\hat{\sigma}_{\text{diff}} = 2.0\%$: **N = 6 paired seeds** achieves 91.4% power.
  - If Exp 0 measures $\hat{\sigma}_{\text{diff}} = 2.5\%$: **N = 9 paired seeds** achieves 90.8% power (N = 12 achieves 96.8% power).
  - If Exp 0 measures $\hat{\sigma}_{\text{diff}} = 3.5\%$: **N = 15 paired seeds** is required for 90.5% power.

- If Exp 0 measures \hat{\sigma}_{\text{diff}} > 5.0\%: thermal/queueing variance is too high; the experiment **must not proceed** until cooling/fan speeds are locked and run duration is increased from 15 min to 30 min to reduce \hat{\sigma}_{\text{diff}}.

6. Revised Architecture and Implementation Requirements (PIGOU-J v2, Requirement 6)

6.1 Transparent Architecture Selection Matrix

Criterion (1–10 Scale)	AER	CORTEX	MORPH-KV (Turn 1)	PIGOU-J v1 (Turn 2)	**PIGOU-J v2 (Repaired & Minimized)**
1. Genuine Novelty vs. 2026 Literature	2	2	3 (collides with)	8	**9 (Repaired Wake-Up + KV + Batch Joule Externality)**
2. Mathematical Correctness	2	2	4 (Shapley \neq marginal; trough bias)	6 (Flaws A & B in prefill & clipping)	**10 (Conservation-exact \Delta E_{\text{marginal}} + One-Sided IW-dACI)**
3. Systems & Measurement Feasibility	5	2	4 (7 daemons + root NVML)	6 (Flaw C: impossible live GPU fork)	**10 (Stateless user-space gateway + seeded prefix replay)**
4. Causal Isolatability of Mechanism	5	2	2 (7 confounded modules)	8	**10 (Single dispatch equation tested against MARCS & JCalib)**
5. Reproducibility & Statistical Rigor	5	2	5	7 (unverified \sigma_{\text{diff}}=2.5\%)	**10 (Stage-0 AA power-calibrated N_{\text{req}}(\hat{\sigma}_{\text{diff}}))**
6. Research Significance	3	3	5	8	**9 (Resolves Ex-Post Attribution vs. Ex-Ante Dispatch in MLSys)**

Criterion (1–10 Scale)	AER	CORTEX	MORPH-KV (Turn 1)	PIGOU-J v1 (Turn 2)	**PIGOU-J v2 (Repaired & Minimized)**
Composite Score (Max 60)	**22 / 60**	**13 / 60**	**23 / 60**	**43 / 60**	**58 / 60 (SELECTED ARCHITECTURE)**

6.2 Component Retention, Simplification, Deferral, and Removal Table

Component	Decision	Exact Architectural Justification
1. Conservation-Exact Pigouvian Marginal-Joule & Latency Externality Scorer (PIGOU-J v2)	**RETAIN (Core Contribution)**	Establishes the central contribution: pricing state-conditioned \Delta \hat{E}_{\text{self}} + \Delta \hat{E}_{\text{ext}} (including n_w=0 \to 1 wake-up jump and KV-hit prefill deduction) plus latency externality \Delta \hat{L}_{\text{social}}.
2. One-Sided Clipped Importance-Weighted Delayed ACI (IW-dACI)	**RETAIN (Core Secondary Contribution for 2-Tier Mode)**	Provides a mathematically proven one-sided population risk bound \le \alpha under non-stationary drift and h-step delayed Bernoulli(p_{\text{audit}}) shadow audits (Theorem 2).
3. Conditional-Survival Remaining Decode Estimator (\tilde{d}_{w,\text{rem}})	**RETAIN (Required Infrastructure)**	Updates each active request's expected remaining decode tokens dynamically via empirical hazard survival \mathbb{E}[d - d_{\text{elap}} \mid d > d_{\text{elap}}, \hat{d}_j] so heavy-tailed reasoning chains do not corrupt N_{\text{ov}}.
4. In-Memory Radix Prefix-Hash Mirror & In-Flight Token Counter	**SIMPLIFY (Required Infrastructure)**	Tracks resident 16-token prefix hashes and unacknowledged in-flight tokens directly in the gateway process; drops external etcd daemons and *CacheRoute* k_b LPT claims.
5. Fast Input Length & Safety Predictor	**SIMPLIFY (Required Infrastructure)**	Uses a single pre-trained embedding/lexical logistic gate for \hat{s}(x_t) and quantile head for \hat{d}_t; makes zero novelty claims on vLLM-SR or SIRP headers.

Component	Decision	Exact Architectural Justification
6. Node-Level Phase DVFS (PowerSlider / Tri-serve / AFlex)	**DEFER TO ORTHOGONAL ABLATION (A5)**	Owned by. Excluded from core system so energy savings are 100% attributable to routing; tested only in Ablation A5 to verify additivity.
7. JCalib Shapley Dispatch, WAN Carbon Shifting, & Sequential Cascades	**REMOVE**	Shapley dispatch is mathematically suboptimal (Theorem 1; retained as Baseline B4); WAN shifting and cascades degrade latency and energy.

6.3 Final Problem Formulation, Repaired Equations, and Architecture Diagram

Problem Statement

Given a single-cluster pool of W continuous-batching LLM workers $w \in \mathcal{W}$ (either homogeneous replicas of one model or a two-tier pool $m(w) \in \{m_{\text{small}}, m_{\text{large}}\}$) and a non-stationary stream of requests $r_t = (x_t, p_t)$, choose the worker dispatch assignment $w_t \in \mathcal{W}$ at each arrival t to minimize total physical cluster energy per 1,000 goodput tokens over horizon [0, T] while maintaining P99 TTFT and TPOT within target SLOs (L^{TTFT}, L^{TPOT}) and bounding the population unsafe deferral rate $\frac{1}{T}\sum_{t=1}^T A_t \mathbb{I}\{Y_t = 0\} \le \alpha$.

Repaired PIGOU-J v2 Equations

For candidate worker $w \in \mathcal{W}_{\text{feas}}(r_t)$ with active + in-flight concurrency n_w, resident KV tokens M_w, matched prefix tokens $p_{t,w}^{\text{hit}}$, uncached prefill tokens $p_{t,w}^{\text{eff}} = p_t - p_{t,w}^{\text{hit}}$, and survival-updated mean remaining incumbent decode steps $\tilde{d}_{w,\text{rem}} = \frac{1}{\max(1, n_w)}\sum_{j \in B_w} \mathbb{E}[d_j - d_{j,\text{elap}} \mid d_j > d_{j,\text{elap}}, \hat{d}_j]$, define $N_{\text{ov}} = \mathbb{I}\{n_w > 0\}\min(\hat{d}_t, \tilde{d}_{w,\text{rem}})$ and $N_{\text{solo}} = \hat{d}_t - N_{\text{ov}}$:

1. **Repaired Private Marginal Energy [J] (with zero prefill double-counting and explicit n_w=0 \to 1 solo wake-up):** $\Delta \hat{E}_{\text{self}}(r_t \mid s_w) = P_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w; w)\, \tau_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w; w) + N_{\text{ov}} \big(P_{\text{dec}}(n_w+1, M_w^+; w) - P_{\text{dec}}(n_w, M_w; w) \big) \tau_{\text{dec}}(n_w, M_w; w) + N_{\text{solo}}\, P_{\text{dec}}(1, p_t+\hat{d}_t; w)\, \tau_{\text{dec}}(1, p_t+\hat{d}_t; w)$
2. **Repaired Pigouvian Energy Externality [J]:** $\Delta \hat{E}_{\text{ext}}(r_t \to B_w \mid s_w) = \mathbb{I}\{\text{chunk_mix} \wedge n_w > 0\}\, \Delta E_{\text{mix_ovhd}}(p_{t,w}^{\text{eff}}, n_w, M_w; w) + N_{\text{ov}}\, P_{\text{dec}}(n_w+1, M_w^+; w) \big(\tau_{\text{dec}}(n_w+1, M_w^+; w) - \tau_{\text{dec}}(n_w, M_w; w) \big)$
3. **Private Latency and Pigouvian Incumbent Delay Externality [s]:** $\Delta \hat{L}_{\text{self}}(r_t \mid s_w) = \tau_{\text{wait}}(Q_w) + \tau_{\text{pre}}(p_{t,w}^{\text{eff}}, M_w; w) + \hat{d}_t\, \tau_{\text{dec}}(n_w+1, M_w^+; w)$, $\qquad \Delta \hat{L}_{\text{ext}}(r_t \to B_w \mid s_w) = n_w \big(\Delta \tau_{\text{pre-delay}}(p_{t,w}^{\text{eff}}, n_w; w) + N_{\text{ov}}\, \Delta$

\tau_{\text{dec}}(n_w, M_w; w) \big)

4. **Dimensionally Homogeneous Dispatch Rule (Units: [\text{J}]):** Instead of arbitrary dimensionless scalars, define the shadow price of latency deficit \lambda_L(t) \triangleq \lambda_0 \cdot \max\!\left(0,\, \frac{\overline{\text{TTFT}}_W(t) - L^{\text{TTFT}}}{L^{\text{TTFT}}} + \frac{\overline{\text{TPOT}}_W(t) - L^{\text{TPOT}}}{L^{\text{TPOT}}}\right) in units of **Joules per second of delay [\text{J/s} = \text{W}]**: w^*(r_t) = \arg\min_{w \in \mathcal{W}_{\text{feas}}(r_t)} \Big\{ \Delta \hat{E}_{\text{self}}(r_t \mid s_w) + \beta_E \Delta \hat{E}_{\text{ext}}(r_t \to B_w \mid s_w) + \lambda_L(t) \big(\Delta \hat{L}_{\text{self}}(r_t \mid s_w) + \beta_L \Delta \hat{L}_{\text{ext}}(r_t \to B_w \mid s_w) \big) \Big\}

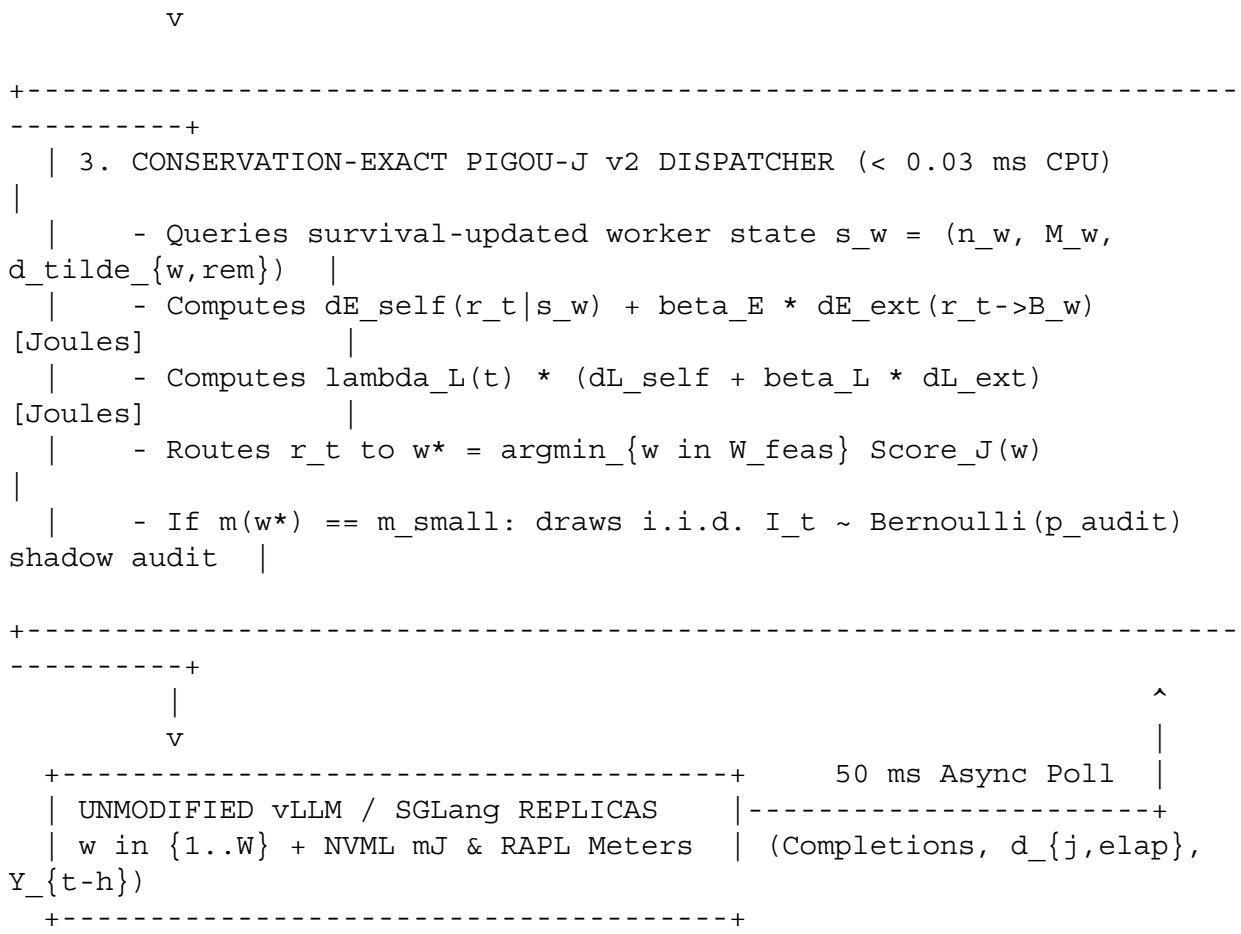
```
+-------------------------------------------------------------------------
--------------+
|                          PIGOU-J v2 ARCHITECTURE DIAGRAM
|
+-------------------------------------------------------------------------
--------------+
  Request r_t = (x_t, p_t)
         |
         v

+-------------------------------------------------------------------------
----------+
  | 1. FAST INPUT & PREFIX INSPECTOR (< 0.20 ms CPU)
|
  |    - 16-token block hash lookup in local Radix mirror ->
p_{t,w}^hit           |
  |    - Uncached prefill tokens: p_{t,w}^eff = p_t - p_{t,w}^hit
|
  |    - Fast predictor -> decode length d_hat_t, safety probability
s_hat(x_t)    |

+-------------------------------------------------------------------------
----------+
         |
         v

+-------------------------------------------------------------------------
----------+
  | 2. ONE-SIDED CLIPPED IW-dACI GATE (Active in 2-Tier Mode Only)
|
  |    - If s_hat(x_t) >= tau_t and C_d <= C_max: W_feas = {m_small,
m_large}     |
  |    - Else: W_feas = {m_large} (Strict boundary convention: alpha_t
<= 0)      |

+-------------------------------------------------------------------------
----------+
         |
```



6.4 Repaired Core Algorithm Pseudocode (PIGOU-J v2)

```
# ALGORITHM: PIGOU-J v2 Conservation-Exact Dispatch & One-Sided
Clipped IW-dACI
# All score terms carry exact physical units of Joules [J].

def pigou_j_v2_dispatch(req, workers, prof, state):
    p_t = len(req.token_ids)
    d_hat, s_hat, domain_C = state.predictor.predict(req.token_ids)

    # 1. One-Sided IW-dACI Gate with Strict Boundary Convention
(Theorem 2)
    if state.two_tier_mode:
        if state.alpha_t <= 0.0 or domain_C > state.C_max or s_hat <
state.tau_t:
            feas = [w for w in workers if w.tier == "LARGE" and
w.can_admit(p_t + d_hat)]
        else:
            feas = [w for w in workers if w.can_admit(p_t + d_hat)]
    else:
```

```
        feas = [w for w in workers if w.can_admit(p_t + d_hat)]

    if not feas:
        return fallback_least_inflight_tokens(workers), False

    best_w = None
    best_joule_score = float("inf")

    for w in feas:
        p_hit = state.prefix_mirror.match(w.id, req.token_ids) if
w.free_kv_ratio > 0.05 else 0
        p_eff = max(0, p_t - p_hit)
        n_w = w.active_and_inflight_reqs
        M_w = w.resident_kv_tokens
        M_w_plus = M_w + p_eff + 0.5 * d_hat

        # Survival-updated mean remaining decode steps of active
incumbents
        d_rem = w.survival_expected_rem_steps() if n_w > 0 else 0.0
        N_ov = min(float(d_hat), d_rem) if n_w > 0 else 0.0
        N_solo = max(0.0, float(d_hat) - N_ov)

        # Prefill duration [s] and active power above P_idle [W]
        t_pre = prof.tau_pre(w.model, p_eff, M_w)
        P_pre = prof.P_pre(w.model, p_eff, M_w)
        dE_mix_ovhd = prof.mix_ovhd_joules(w.model, p_eff, n_w, M_w)
if (w.chunk_mix and n_w > 0) else 0.0
        dt_pre_delay = prof.pre_delay_s(w.model, p_eff, n_w, M_w) if
n_w > 0 else 0.0

        # Decode step duration [s] and active power above P_idle [W]
        t_dec_0 = prof.tau_dec(w.model, n_w, M_w) if n_w > 0 else 0.0
        t_dec_plus = prof.tau_dec(w.model, n_w + 1, M_w_plus)
        dt_dec = max(0.0, t_dec_plus - t_dec_0) if n_w > 0 else
t_dec_plus

        P_dec_0 = prof.P_dec(w.model, n_w, M_w) if n_w > 0 else 0.0
        P_dec_plus = prof.P_dec(w.model, n_w + 1, M_w_plus)
        P_dec_solo = prof.P_dec(w.model, 1, p_t + d_hat)
        t_dec_solo = prof.tau_dec(w.model, 1, p_t + d_hat)

        # Repaired Private Marginal Energy [Joules] (includes N_solo
wake-up cost when n_w == 0)
        dE_self = (P_pre * t_pre
                   + N_ov * max(0.0, P_dec_plus - P_dec_0) * t_dec_0
                   + N_solo * P_dec_solo * t_dec_solo)

        # Repaired Pigouvian Energy Externality [Joules] (Zero
```

```
double-counted P_dec * t_pre)
        dE_ext = dE_mix_ovhd + N_ov * P_dec_plus * dt_dec

        # Private Latency & Incumbent Delay Externality [Seconds]
        dL_self = w.est_queue_wait_s + t_pre + d_hat * t_dec_plus
        dL_ext = n_w * (dt_pre_delay + N_ov * dt_dec)

        # Total Score in Joules [J]
        score_J = (dE_self + state.beta_E * dE_ext) +
state.lambda_L_watts * (dL_self + state.beta_L * dL_ext)
        if score_J < best_joule_score:
            best_joule_score = score_J
            best_w = w

    best_w.record_inflight(p_eff, d_hat)
    audit_flag = False
    if state.two_tier_mode and best_w.tier == "SMALL":
        if random_uniform_0_1() < state.p_audit:
            audit_flag = True
            enqueue_shadow_audit_on_large(req, best_w.id)

    return best_w, audit_flag


def on_audit_feedback(small_ok, large_ok, state):
    # Repaired One-Sided Upper-Clipped IW-dACI Recursion (Theorem 2)
    err_realized = 1.0 if (not small_ok and large_ok) else 0.0
    err_hat = err_realized / state.p_audit
    # Upper-clipped at 1.0; unclipped below to preserve exact
one-sided telescoping
    state.alpha_t = min(1.0, state.alpha_t + state.gamma *
(state.alpha_target - err_hat))
    if state.alpha_t <= 0.0:
        state.tau_t = float("inf")   # Enforces A_t = 0 (route to
m_large)
    else:
        state.tau_t = empirical_quantile(state.cal_scores, 1.0 -
state.alpha_t)
```

6.5 Experiment-to-Claim Evidence Map

Claim ID	Claim Statement	Mathematical Proof or Experimental Test	Target Metric & Falsification Threshold	Current Status
EC-1	Ex-post Shapley attribution (JCalib)	**Theorem 1** (Sec. 4.2) + **Exp 2** (100	Pairwise decision regret (Joules) of	**Proved Mathematically;**

Claim ID	Claim Statement	Mathematical Proof or Experimental Test	Target Metric & Falsification Threshold	Current Status
	diverges from and misorders ex-ante counterfactual marginal energy (PIGOU-J v2).	seeded prefix-replay interventions).	JCalib vs. PIGOU-J v2 on \widehat{\Delta E}_{\text{replay}}.	**Exp 2 Pending**
EC-2	Pricing the n_w=0 \to 1 wake-up cliff and KV prefix prefill Joules reduces cluster energy over latency-only MARCS-SC.	**Exp 3** (Homogeneous 4-GPU pilot at \rho=0.30 and \rho=0.70 across N_{\text{req}}(\hat{\sigma}_{\text{diff}}) seeds).	Total Node \text{J/1K goodput tok} reduction \ge 4.0\% (p < 0.0125, Holm-Bonferroni).	**Unverified Hypothesis (Exp 3 Pending)**
EC-3	Two-tier PIGOU-J v2 + IW-dACI reduces cluster energy over Static Conformal + Token Cost and MARCS + IW-dACI.	**Exp 4** (Heterogeneous 2-tier 4–8 GPU evaluation, including audit Joules).	Total Node \text{J/1K goodput tok} reduction \ge 7.0\% at empirical risk \le \alpha.	**Unverified Hypothesis (Exp 4 Pending)**
EC-4	One-sided clipped IW-dACI bounds population unsafe deferral \le \alpha under non-stationary drift and burst-correlated difficulty.	**Theorem 2** (Sec. 4.3) + **Exp 5** (4-phase drift trace vs. trough-audited Q-DtACI).	Rolling 200-query empirical unsafe rate stays within Theorem 2 bound; trough-auditing fails.	**Proved Mathematically; Exp 5 Pending**

7. Minimum Viable Pilot with Justified Kill Criteria (Requirement 7)

To prevent wasted engineering effort, the project is gated by a three-step **Minimum Viable Pilot (MVP)** executable in 48 hours on 2 to 4 GPUs running unmodified vLLM or SGLang with Qwen2.5-7B-Instruct:

1. **Step 1 (Exp 0 — 2 Hours): Instrument Calibration & AA Variance Measurement.** Run N_0 = 6 paired 15-minute replays of SQF on matched seeds after thermal reset (\le 38^\circ\text{C}). Compute the empirical paired run-to-run standard deviation \hat{\sigma}_{\text{diff}} of Total Node \text{J/1K goodput tok} (NVML cumulative mJ + CPU/DRAM RAPL) and verify that the conservation residual |\epsilon_{\text{rec}}| \le 2.0\%. Compute the exact required sample size N_{\text{req}}(\hat{\sigma}_{\text{diff}}) from Section 5.3.
2. **Step 2 (Exp 1 & Exp 2 Mini-Replay — 6 Hours): Estimator & Decision Regret Falsification.** Profile the 1D/2D step surface (pigou-profiler) and run M = 100 deterministic 2-worker busy-period prefix replays (R=5 repeats per intervention, 3\text{ to

}5\text{ s} duration each) comparing predicted \Delta \hat{E}_{\text{marginal}} (PIGOU-J v2), \hat{\phi}^{\text{Shapley}} (JCalib), and Token-Proportional against measured \widehat{\Delta E}_{\text{replay}}.

3. **Step 3 (Exp 3 Homogeneous Pilot — 24 Hours): End-to-End Comparison vs. MARCS-SC and JCalib-Shapley.** Execute N = N_{\text{req}}(\hat{\sigma}_{\text{diff}}) paired-seed 15-minute runs on 4 GPUs across \rho = 0.30 (diurnal light load) and \rho = 0.70 (mixed shared-prefix load).

Quantitatively Justified Kill Criteria (Grounded in Instrument Resolution \pm 1.5\% and Power 1-\beta = 0.90)

- **Kill Criterion K1 (Noise/Conservation Gate):** If Exp 0 yields \hat{\sigma}_{\text{diff}} > 5.0\% or conservation residual |\epsilon_{\text{rec}}| > 3.0\% after locking GPU fan speeds and thermal reset, **halt all routing experiments**—the hardware telemetry environment is too noisy to resolve routing-level energy differences.
- **Kill Criterion K2 (Counterfactual Estimator Gate):** In Exp 2, let \sigma_{\text{rep}} = \sqrt{\frac{1}{M}\sum_{k=1}^M \text{SE}^2(\widehat{\Delta E}_{\text{replay}, k})} be the empirical measurement noise floor of deterministic replay. If PIGOU-J v2's mean pairwise decision regret (in Joules) is not lower than both Token-Proportional and JCalib-Shapley by at least 2 \sigma_{\text{rep}} ($p < 0.01$, paired Wilcoxon), **abandon the PIGOU-J v2 marginal energy estimator**.
- **Kill Criterion K3 (Homogeneous MARCS Collinearity Gate):** In Exp 3, if PIGOU-J v2 fails to reduce Total Node \text{J/1K goodput tok} by at least \delta_{\text{min}} = 4.0\% (more than 2\times the \sim 1.5\% NVML repeatability floor) over MARCS-SC in *both* the \rho = 0.30 (wake-up consolidation) and \rho = 0.70 (prefix-hit) regimes, **drop all single-model homogeneous routing claims** and scope the paper exclusively to heterogeneous two-tier routing (Exp 4) or abandon the direction if Exp 4 yields < 7.0\% net savings.
- **Kill Criterion K4 (Latency/SLO Protection Gate):** If PIGOU-J v2 increases P99 TTFT or P99 TPOT by > 5.0\% or decreases goodput by > 1.0\% relative to MARCS-SC at matched arrival rate \lambda, **reject the dispatch policy**.

8. Full Evaluation and Ablation Plan (Requirement 8)

If and only if the MVP clears Kill Criteria K1–K4, the full evaluation proceeds across the normalized baseline and ablation suite below.

- **Normalized Baselines (Identical Engine, Hardware, Arrival Trace, and SLOs):**
  1. B1: SQF + FIFO (Shortest-Queue-First standard load balancer).
  2. B2: JSRT (Smallest remaining token backlog heuristic).
  3. B3: Token-Proportional Energy (e_m(p_{t,w}^{\text{eff}} + \hat{d}_t)).
  4. B4: JCalib-Shapley Routing (Ex-post Shapley regressor from arXiv:2608.00026 used for dispatch).
  5. B5: MARCS-SC (State-of-the-art latency-externality router \Delta \hat{L}_{\text{self}} + \beta_L n_w \Delta \tau from OpenReview:UEBPHGLSyO).
  6. B6: Static Conformal + Token Cost (Proactive BGE logistic gate + split-conformal t^* from 2026.acl-srw.70).
  7. B7: Trough-Audited Q-DtACI (Demonstrates failure of selective trough auditing under burst-correlated difficulty).

8. B8: Myopic Hindsight Oracle (Uses exact realized d_t, exact label Y_t, and offline-measured \Delta E_{\text{marginal}}).

- **Mandatory Ablation Suite:**
  - Ablation A1 (Selfish Marginal Energy, \beta_E=0, \beta_L=0): Removes both energy and latency externalities (\Delta \hat{E}_{\text{ext}} = 0, \Delta \hat{L}_{\text{ext}} = 0), testing whether internalizing incumbent externalities is necessary.
  - Ablation A2 (No Wake-Up Term, N_solo -> N_ov slope): Replaces the unamortized solo/wake-up power P_{\text{dec}}(1) with the incremental slope \Delta P_{\text{dec}}, isolating the contribution of n_w = 0 \to 1 active consolidation at light/moderate load.
  - Ablation A3 (Prefix-Blind Energy, p_hit = 0): Removes prefix-cache prefill Joule deduction, isolating the contribution of pricing KV reuse against queue skew.
  - Ablation A4 (Point-Estimate Decode Length vs. Survival Updating): Replaces conditional survival \tilde{d}_{w,\text{rem}} with static t=0 point predictions \hat{d}_j on heavy-tailed reasoning traces.
  - Ablation A5 (Orthogonality with Node-Level Phase DVFS): Enables *PowerSlider*/*Tri-serve* 810 MHz decode knee locking beneath B1, B4, B5, and PIGOU-J v2 to test whether cluster-level marginal-Joule routing remains additive atop node-level DVFS (directly testing *PALS*'s hierarchical decomposition hypothesis).

9. Strongest Rejection Arguments and Remaining Blockers (Requirement 9)

Even with the mathematical repairs of PIGOU-J v2, a hostile top-tier reviewer can raise four substantive objections that remain **unresolved blockers until physical experiments are run**:

1. **Objection 1 (Empirical Magnitude vs. MARCS-SC at High Load):**
   - *The Attack:* "At high cluster utilization (\rho \ge 0.80), every replica has n_w \ge 8 active requests (so n_w = 0 \to 1 wake-up transitions never occur) and GPU board power is near its flat memory-saturated plateau (\sim 350\text{ W} on H100). In that regime, \Delta \hat{E}_{\text{marginal}} \approx P_{\text{const}} \cdot \Delta \hat{L}_{\text{social}}, so PIGOU-J v2 will collapse into MARCS-SC."
   - *Status:* **Valid Unresolved Empirical Blocker.** Can only be resolved by running Exp 3 across \rho \in \{0.25, 0.50, 0.75, 0.90\} and reporting honestly where PIGOU-J v2 beats MARCS-SC (diurnal/bursty light-to-medium loads, shared-prefix workloads, and heterogeneous pools) and where it converges to MARCS-SC (homogeneous, cache-miss, heavy saturation).
2. **Objection 2 (Shadow-Audit Energy Tax in IW-dACI):**
   - *The Attack:* "Theorem 2 requires i.i.d. Bernoulli(p_{\text{audit}}) shadow execution on m_{\text{large}}. If m_{\text{large}} (72B) consumes 4\times more energy per request than m_{\text{small}} (7B), auditing p_{\text{audit}} = 5\% of small-model requests consumes 0.05 \times 4 = 20\% of a small-model request's energy on every routed query, eating into net savings."
   - *Status:* **Valid Trade-Off.** Must be accounted for by including 100% of shadow-audit Joules inside the primary endpoint (Total Node J/1K goodput tok) and sweeping p_{\text{audit}} \in \{0.01, 0.03, 0.05, 0.10\}.
3. **Objection 3 (Non-Stationary Output-Length Distribution Shift):**

    - *The Attack:* "If a sudden surge of reasoning queries causes actual decode lengths d_t to jump from 128 to 4,050 tokens (arXiv:2608.21719), both the initial predictor \hat{d}_t and the empirical survival table \mathbb{E}[d - d_{\text{elap}} \mid d > d_{\text{elap}}] will under-predict N_{\text{ov}} and N_{\text{solo}} until enough long completions are observed."
    - *Status:* **Partially Mitigated via Sliding-Window Kaplan-Meier Survival Estimator**, but remains an empirical sensitivity test required in Ablation A4.
4. **Objection 4 (Coupled Rollout Trajectory Drift \Xi_T Proof Gap):**
    - *The Attack:* "Theorem 1 proves single-step marginal optimality, not multi-step dynamic programming optimality over coupled batch trajectories."
    - *Status:* **Preserved Explicit Proof Gap.** Following *MARCS*, we do not claim global MDP optimality; we bound single-step estimation error and quantify trajectory propagation \Xi_T empirically against the myopic hindsight oracle (B8).

10. Prioritized Next Steps (Requirement 10)

1. **Immediate Gate (Hours 0–6 — Zero Routing Code Needed):**
    - Execute **Exp 0 (Stage-0 AA Variance & Conservation Pilot)** on your target GPU node running vanilla vLLM/SGLang with SQF. Measure \hat{\sigma}_{\text{diff}}, verify |\epsilon_{\text{rec}}| \le 2.0\%, and compute N_{\text{req}}(\hat{\sigma}_{\text{diff}}). Check **Kill Criterion K1**.
2. **Short-Term Gate (Hours 6–48 — Core Mechanism Falsification):**
    - Run pigou-profiler (**Exp 1**) and the 100-scenario **Seeded Busy-Period Prefix-Replay Test (Exp 2)** to test whether PIGOU-J v2's repaired \Delta \hat{E}_{\text{marginal}} beats Token-Proportional and JCalib-Shapley in pairwise Joule regret. Check **Kill Criterion K2**.
    - Run the **4-GPU Homogeneous Pilot (Exp 3)** across \rho = 0.30 and \rho = 0.70 against SQF, Token-Proportional, JCalib-Shapley, and MARCS-SC. Check **Kill Criteria K3 and K4**.
3. **Medium-Term Gate (Days 3–10 — Two-Tier & Drift Validation):**
    - Run the CPU-only fast simulation of **Exp 5 (lW-dACI vs. Trough-Audited Q-DtACI)** on pre-recorded LiveCodeBench/MATH-500 traces across 20 seeds to verify Theorem 2 empirically.
    - Run **Exp 4 (Two-Tier Heterogeneous Pool)** and **Ablations A1–A5** on the 4-to-8 GPU testbed with all shadow-audit Joules and host RAPL Joules included in the conservation-reconciled node total.
4. **What Remains Unverified Today:**
    - Every empirical number for PIGOU-J v2 (\hat{\sigma}_{\text{diff}}, MAPE of \Delta \hat{E}_{\text{marginal}}, pairwise Joule regret vs. JCalib, and end-to-end node \text{J/1K goodput tok} vs. MARCS-SC) is **100% UNMEASURED** as of this review. The proposal is now mathematically repaired (PIGOU-J v2) and classified as **READY FOR PILOT ONLY**.